

Project Report: Gamified Clinical Assessment Module

Feature: Sliding Tile Puzzle for "Loss of Interest" Evaluation

1. Executive Summary

As part of our psychological support software, we are developing a gamified assessment tool designed to help psychiatrists evaluate patients in a non-intimidating, friendly environment. By integrating a customizable sliding tile puzzle that is openly accessible within the app, our software allows patients to engage in a relaxing activity while providing clinicians with a secure, backend dashboard to review cognitive and motor function data. This module specifically targets "Loss of Interest" (anhedonia) while maintaining the comforting, "Mental Friend" (মনের বন্ধু) persona of our platform.

2. Core Concept

The feature centers around a digital sliding tile puzzle. The puzzle's image and difficulty can be tailored on the spot using a simple tag-based search. This allows the patient to choose something they like (or the psychiatrist to guide the choice), ensuring maximum engagement. The application records the background data of each session, turning a simple mini-game into a subtle, interactive diagnostic tool.

3. User Experience & Clinical Workflow

The feature is designed for seamless, barrier-free access, functioning effectively whether the patient is alone or in a live session with their psychiatrist:

- **Step 1: Universal Access & Configuration:** The interface features a highly visible, simple "Start a Game" button accessible to anyone at any time. Once tapped, the user (patient or psychiatrist) is presented with quick configuration options. They can adjust the difficulty (Easy, Medium, Hard), tweak parameters (time and move limits), and select an image. Hitting "Enter" immediately starts the game.
- **Step 2: Stress-Free Gameplay:** The patient interacts with a clean interface. If used during a clinical session, the psychiatrist can physically observe the patient's reactions, physical tension, and engagement levels while they play.
- **Step 3: Gentle Completion & Reset:** Once the puzzle ends, whether completed, abandoned or exited, the screen displays a friendly "Thank you for Playing" message. A simple tap on the screen instantly resets the interface back to the "Start a Game" screen, ensuring there are no intimidating "Game Over" or "Failed" screens.

- **Step 4: Secure Clinical Dashboard:** On the top right corner of the main screen, there is a discrete "Login" button. The psychiatrist can enter a simple PIN to access the secure backend. Here, every puzzle session is automatically logged and sorted by date and time (latest to oldest). The psychiatrist can inspect individual sessions to view hidden metrics like total time spent, number of moves, and completion status.

4. Clinical Value & MSE Integration

This self-service mini-game directly supports the Mental Status Examination (MSE) outlined in our foundational survey:

- **Measuring "Loss of Interest":** Giving the patient the freedom to use the tag-based search is an assessment in itself. If they search for a topic they usually love but still show no motivation to play, or if they cannot think of a single tag to search, it provides an objective indicator of anhedonia.
- **Evaluating Motor Activities:** The physical act of sliding tiles allows the clinician to observe psychomotor retardation or agitation during in-person sessions.
- **Assessing Cognition:** The secure dashboard metrics (moves made, time taken) provide hard data on attention, concentration, and decision-making capacity without making the patient feel tested.
- **Anxiety Mitigation:** By keeping the setup open and the completion screen positive ("Thank you for Playing"), we prevent the game from becoming a source of performance anxiety.

5. Future Roadmap: The "Hint" System

In future iterations, we plan to introduce a standard algorithm-driven "Hint" button during gameplay that shows the optimal next sliding move. Clinically, this feature will allow the psychiatrist to measure:

- **Frustration Tolerance:** Observing how quickly a patient gives up versus asking for help.
- **Confidence:** Noting if the patient becomes overly reliant on the hint system rather than trusting their own problem-solving skills.

6. Conclusion

The sliding tile puzzle module perfectly bridges the gap between patient autonomy and clinical assessment. By making the game universally accessible and customizable, we empower the patient to engage with the app as a true "companion." Simultaneously, the discrete PIN-locked dashboard ensures our psychiatrists have instant access to organized, chronological data to aid in their vital diagnostic work.